

PROCÉDURE TECHNIQUE

Haute Disponibilité Web

Apache2 & Heartbeat

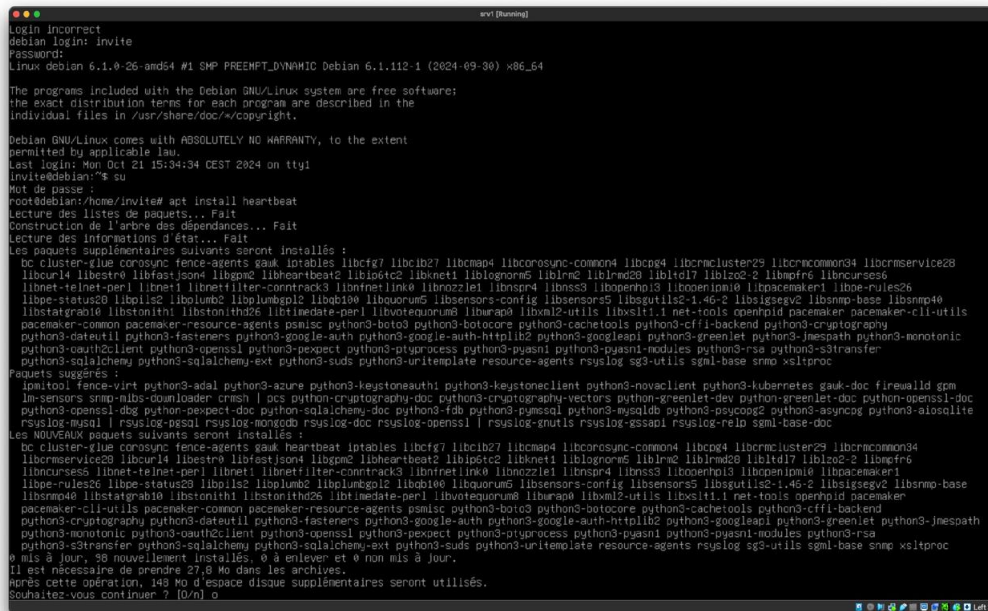
Infrastructure Web redondée sur Debian — Continuité de service

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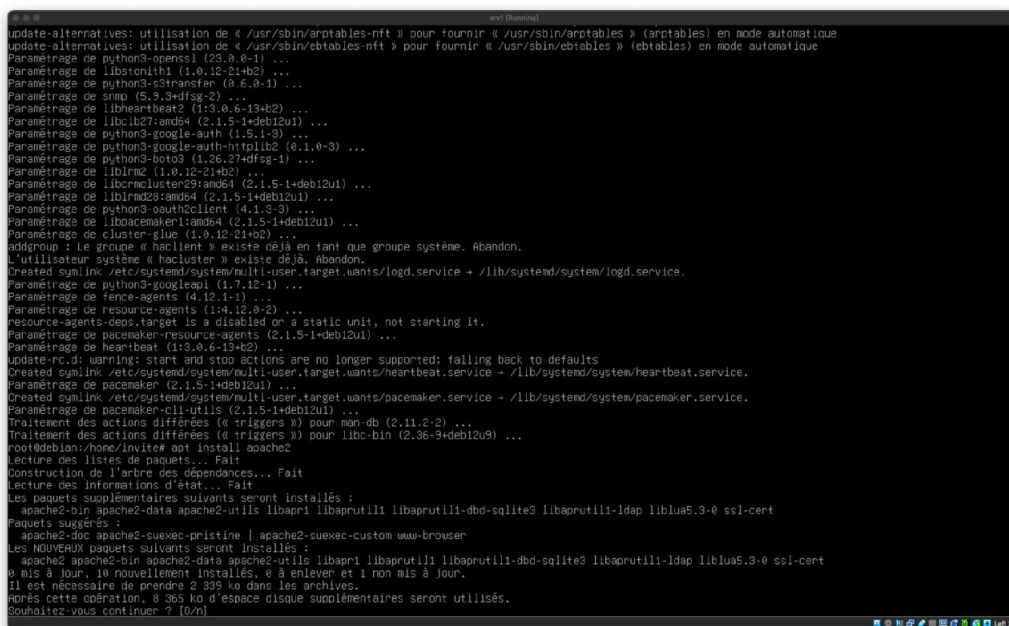
Document confidentiel — Usage interne

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Installation de apache2



Changement du nom du host

```

traitement des actions différées (« triggers ») pour libc-bin (2.36
root@debian:/home/invite# hostnamectl set-hostname srvWeb1
root@debian:/home/invite# _

```

Configuration de la carte réseau

```

# This file describes the network interfaces available on your syst
# and how to activate them. For more information, see interfaces(5)

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug enp0s3
iface enp0s3 inet dhcp

auto enp0s8
iface enp0s8 inet static
    address 172.20.0.11
    netmask 255.255.255.0_

```

La commande pour pouvoir modifier le config ip

```

root@srvWeb1:/home/invite#
root@srvWeb1:/home/invite#
root@srvWeb1:/home/invite# nano /etc/network/interfaces

```

Téléchargement d'un template pour le apache server web

```

root@srvWeb1:/home/invite# wget https://github.com/technext/thegrill/archive/master.
--2025-03-04 15:02:24-- https://github.com/technext/thegrill/archive/master.zip
Résolution de github.com (github.com)... 140.82.121.4
Connexion à github.com (github.com)[140.82.121.4]:443... connecté.
requête HTTP transmise, en attente de la réponse... 302 Found
Emplacement : https://codeload.github.com/technext/thegrill/zip/refs/heads/master [s
--2025-03-04 15:02:24-- https://codeload.github.com/technext/thegrill/zip/refs/head
Résolution de codeload.github.com (codeload.github.com)... 140.82.121.10
Connexion à codeload.github.com (codeload.github.com)[140.82.121.10]:443... connecté.
requête HTTP transmise, en attente de la réponse... 200 OK
Taille : non indiqué [application/zip]
Sauvegarde en : « master.zip »

```

```

master.zip [ <=>

```

```

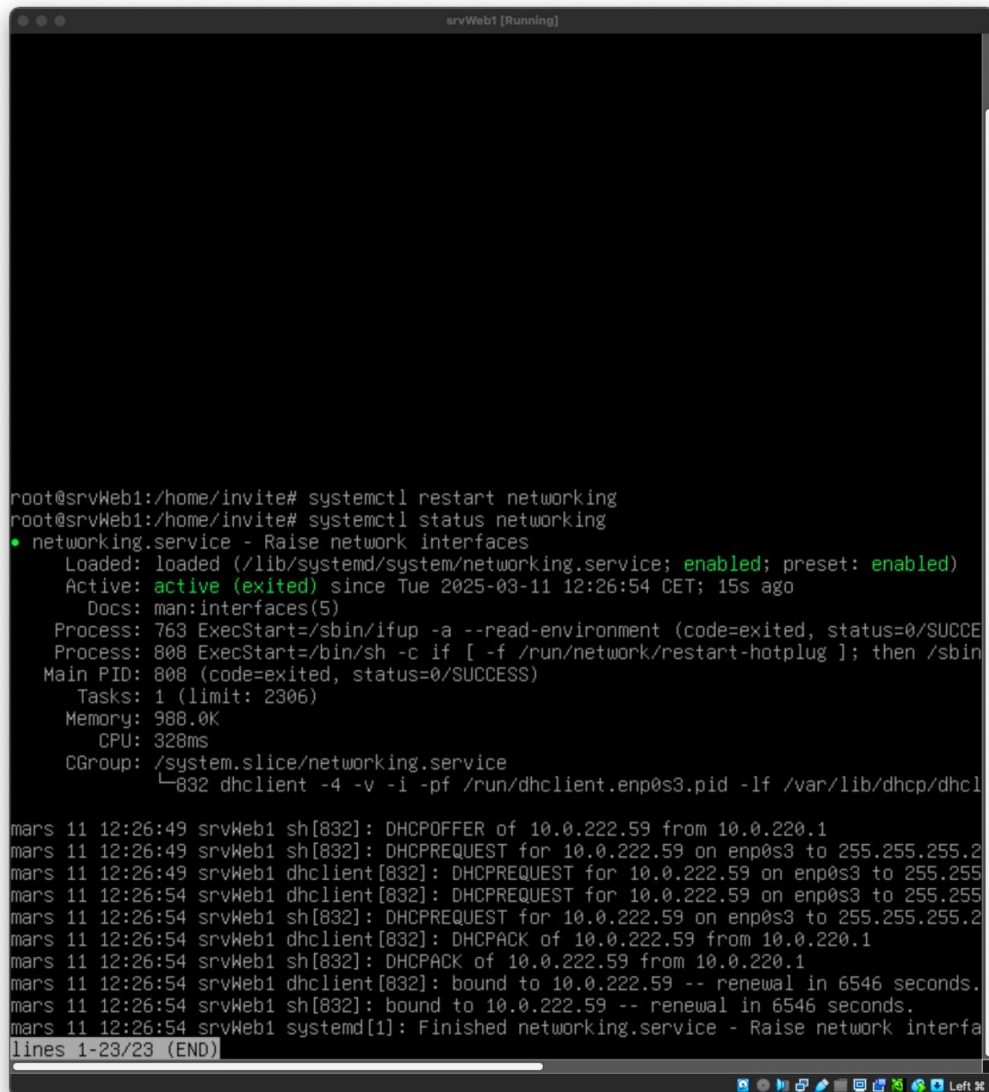
2025-03-04 15:02:26 (2,94 MB/s) - « master.zip » sauvegardé [4534581]

```

Dezip

```
creating: thegrill-master/assets/img/ico/
inflating: thegrill-master/assets/img/ico/apple-touch-icon-114x114.png
inflating: thegrill-master/assets/img/ico/apple-touch-icon-144x144.png
extracting: thegrill-master/assets/img/ico/apple-touch-icon-57x57.png
extracting: thegrill-master/assets/img/ico/apple-touch-icon-72x72.png
inflating: thegrill-master/assets/img/ico/favicon.ico
inflating: thegrill-master/assets/img/intro-logo.jpg
inflating: thegrill-master/assets/img/intro-logo.png
extracting: thegrill-master/assets/img/large_logo.png
inflating: thegrill-master/assets/img/loading.GIF
creating: thegrill-master/assets/img/logo/
inflating: thegrill-master/assets/img/logo/logo1.png
inflating: thegrill-master/assets/img/logo/logo2.png
inflating: thegrill-master/assets/img/logo/logo3.png
inflating: thegrill-master/assets/img/logo/logo4.png
inflating: thegrill-master/assets/img/logo/logo5.png
inflating: thegrill-master/assets/img/logo/logo6.png
inflating: thegrill-master/assets/img/logo/logo7.png
inflating: thegrill-master/assets/img/logo/logo8.png
inflating: thegrill-master/assets/img/map-image.png
inflating: thegrill-master/assets/img/marker.GIF
inflating: thegrill-master/assets/img/marker.png
creating: thegrill-master/assets/img/team/
inflating: thegrill-master/assets/img/team/1.jpg
inflating: thegrill-master/assets/img/team/2.jpg
inflating: thegrill-master/assets/img/team/3.jpg
inflating: thegrill-master/assets/img/team/4.jpg
inflating: thegrill-master/assets/img/team/small1.jpg
inflating: thegrill-master/assets/img/team/small2.jpg
inflating: thegrill-master/assets/img/team/team1.jpg
inflating: thegrill-master/assets/img/team/team2.jpg
extracting: thegrill-master/assets/img/world-map.png
creating: thegrill-master/assets/js/
inflating: thegrill-master/assets/js/bigvideo.js
inflating: thegrill-master/assets/js/bootstrap.js
inflating: thegrill-master/assets/js/bootstrap.min.js
inflating: thegrill-master/assets/js/init.js
inflating: thegrill-master/assets/js/jquery.js
inflating: thegrill-master/assets/js/plugins.js
inflating: thegrill-master/index-video.html
inflating: thegrill-master/index.html
inflating: thegrill-master/single-post.html
root@srvWeb1:/home/invite# ls
master.zip thegrill-master
root@srvWeb1:/home/invite# unzip master.zip
```

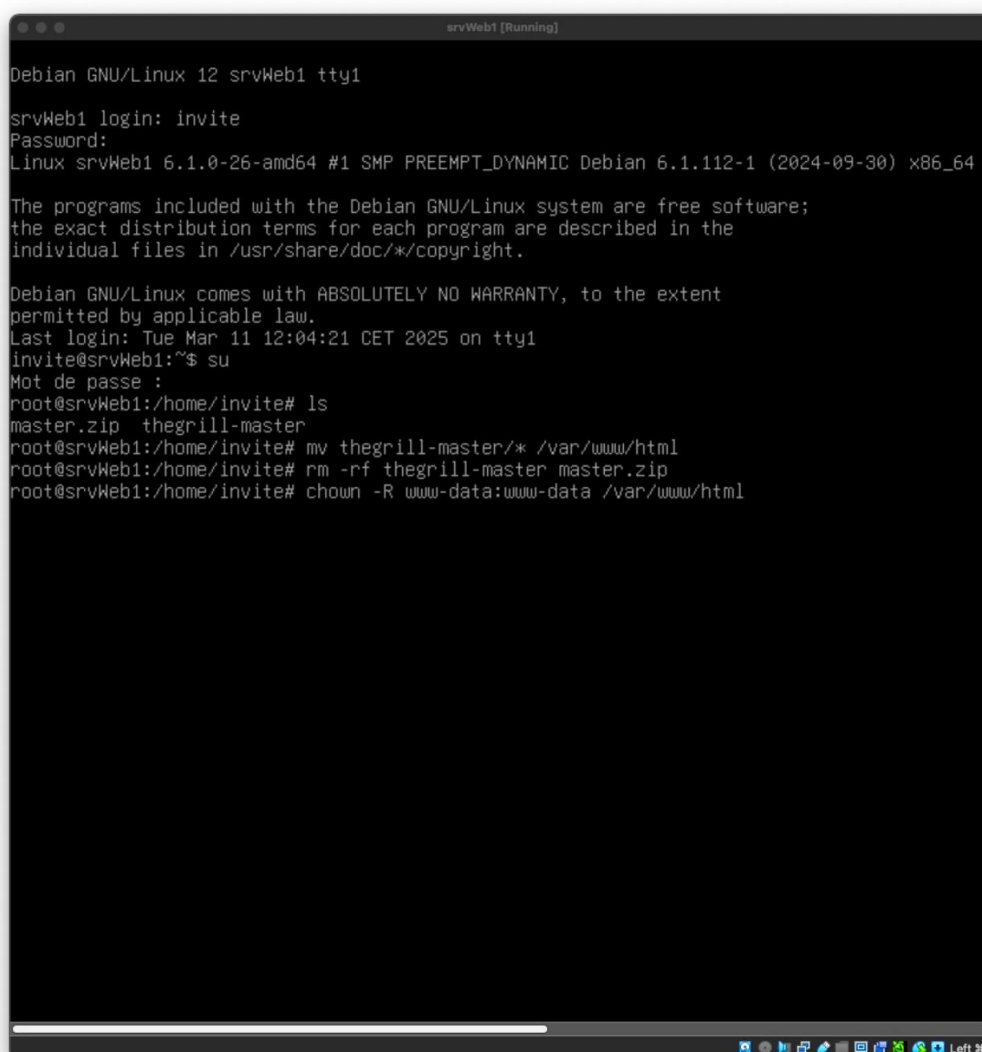
Verification des interfaces

A terminal window titled 'srvWeb1 [Running]' showing a sequence of commands and their outputs. The user runs 'systemctl restart networking' and 'systemctl status networking'. The status output shows the service is active (exited) and provides details about its configuration and resources. Following this, a series of DHCP-related log messages are displayed, showing the process of requesting and receiving an IP address (10.0.222.59) from a DHCP server (10.0.220.1) on the enp0s3 interface. The logs include timestamps and process IDs for each step, from the initial offer to the final binding and renewal information.

```
root@srvWeb1:/home/invite# systemctl restart networking
root@srvWeb1:/home/invite# systemctl status networking
• networking.service - Raise network interfaces
   Loaded: loaded (/lib/systemd/system/networking.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2025-03-11 12:26:54 CET; 15s ago
     Docs: man:interfaces(5)
  Process: 763 ExecStart=/sbin/ifup -a --read-environment (code=exited, status=0/SUCCESS)
  Process: 808 ExecStart=/bin/sh -c if [ -f /run/network/restart-hotplug ]; then /sbin
 Main PID: 808 (code=exited, status=0/SUCCESS)
    Tasks: 1 (limit: 2306)
   Memory: 988.0K
      CPU: 328ms
   CGroup: /system.slice/networking.service
           └─832 dhclient -4 -v -i -pf /run/dhclient.enp0s3.pid -lf /var/lib/dhcp/dhcl

mars 11 12:26:49 srvWeb1 sh[832]: DHCPOFFER of 10.0.222.59 from 10.0.220.1
mars 11 12:26:49 srvWeb1 sh[832]: DHCPREQUEST for 10.0.222.59 on enp0s3 to 255.255.255.2
mars 11 12:26:49 srvWeb1 dhclient[832]: DHCPREQUEST for 10.0.222.59 on enp0s3 to 255.255
mars 11 12:26:54 srvWeb1 sh[832]: DHCPREQUEST for 10.0.222.59 on enp0s3 to 255.255
mars 11 12:26:54 srvWeb1 dhclient[832]: DHCPACK of 10.0.222.59 from 10.0.220.1
mars 11 12:26:54 srvWeb1 sh[832]: DHCPACK of 10.0.222.59 from 10.0.220.1
mars 11 12:26:54 srvWeb1 dhclient[832]: bound to 10.0.222.59 -- renewal in 6546 seconds.
mars 11 12:26:54 srvWeb1 sh[832]: bound to 10.0.222.59 -- renewal in 6546 seconds.
mars 11 12:26:54 srvWeb1 systemd[1]: Finished networking.service - Raise network interfa
lines 1-23/23 (END)
```

deplace -ment du template



```
Debian GNU/Linux 12 srvWeb1 tty1

srvWeb1 login: invite
Password:
Linux srvWeb1 6.1.0-26-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.112-1 (2024-09-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Mar 11 12:04:21 CET 2025 on tty1
invite@srvWeb1:~$ su
Mot de passe :
root@srvWeb1:/home/invite# ls
master.zip  thegrill-master
root@srvWeb1:/home/invite# mv thegrill-master/* /var/www/html
root@srvWeb1:/home/invite# rm -rf thegrill-master master.zip
root@srvWeb1:/home/invite# chown -R www-data:www-data /var/www/html
```

config de apache


```
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

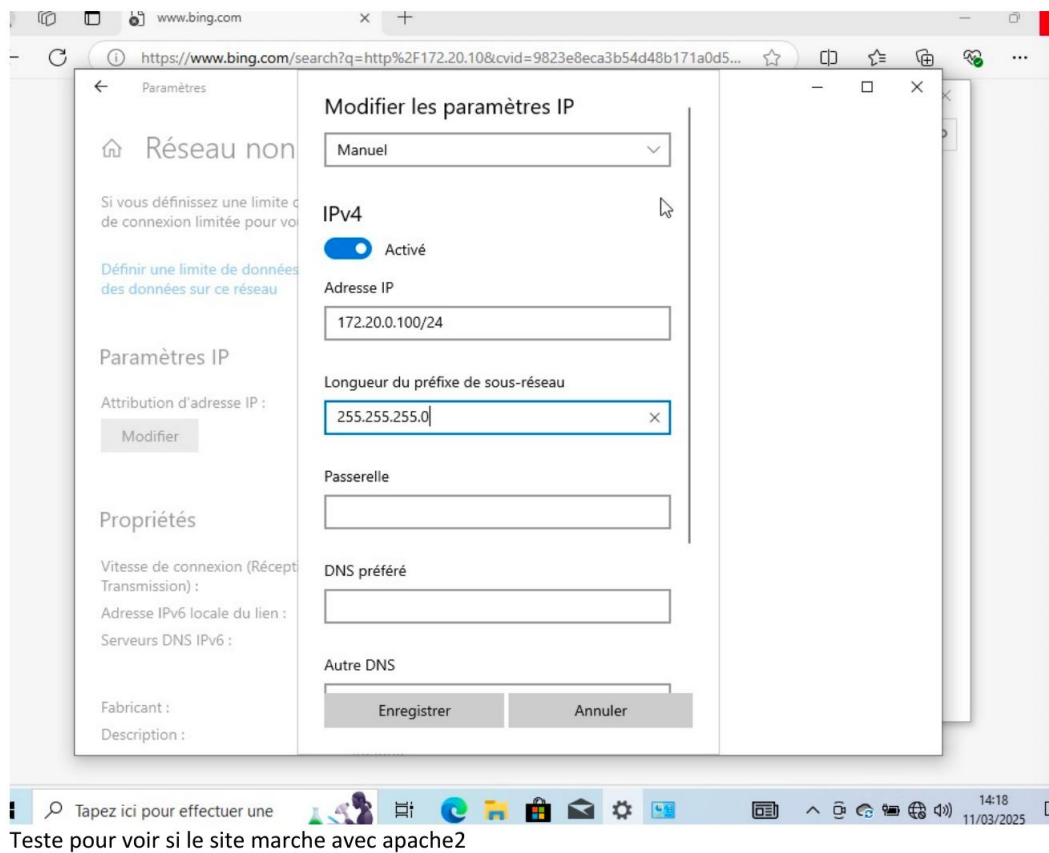
ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

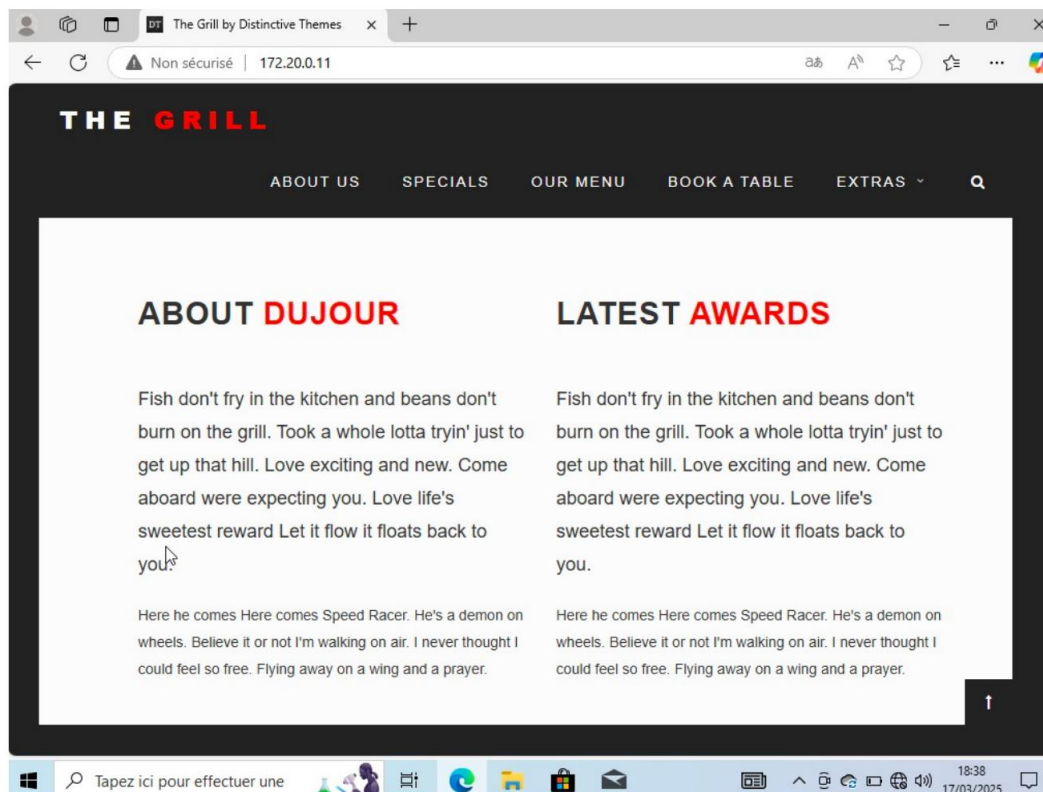
# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
```

Vérification de la configuration de apache2

```
root@srvWeb1:/home/invite# systemctl restart apache2
```

Config de l'ip du client window





Configuration de heartbeat



Mise en place du premier srvweb prioritaire

```
GNU nano 7.2 /etc/ha.d/haresources
srvWeb1 IPaddr::172.20.0.10/24/enp0s8 apache2
```

mise ne place de l'authenrification

```
GNU nano 7.2 /etc/ha.d/authkeys *
auth 1
1 md5 greta
```

Protection du fichier d'auth pour élever son niveau d'accès

```
root@srvWeb1:/home/invite# chmod 600 /etc/ha.d/authkeys _
```

on desactive apache2 au démarrage

```
root@srvWeb1:/home/invite# chmod 600 /etc/ha.d/authkeys
root@srvWeb1:/home/invite# systemctl disable apache2_
```